Acme Games Secure Network Lab

What this lab is

Acme Games is a fictional 65-person video game studio that runs its own online game servers. This lab rebuilds a small version of its network using virtual machines, designs it to resist attack, and tests whether it holds up. A script was written to simulate the dedicated video game server.

How it is built

The network is split into four separate zones. A firewall acts as the guard at every door and only lets traffic through when there is an explicit reason.

- User zone: staff computers.
- Server zone: the main server, which handles sign-ins and stores the studio's code, game builds and artwork.
- Management zone: the IT team's computer, which runs everything and keeps the security records.
- DMZ: It is the only thing the internet can reach, and it is sealed off from everything else. The game server players connect to lives here.
- Simulated internet: a computer outside the firewall, used to test what outsiders can see.

Main protections. People only get the access their job needs: artists cannot see the source code. Passwords are long, and accounts lock after five wrong guesses. The most powerful admin accounts cannot sign in to staff computers. Security records are copied to the management zone, out of an attacker's reach. Shared files have automatic restore points. The game server runs with minimal permissions, ignores junk traffic and cannot start connections to anything.

Problems encountered and how they were resolved

Problem	Resolution
The firewall's control page could not be reached, even though the firewall seemed to answer.	The host PC had accidentally been given the firewall's own address. Removing the PC from the lab networks fixed it and closed a hidden back door.
The firewall's connections were plugged into the wrong zones.	The virtual "cables" were moved to the correct zones.
Security settings sat in one catch-all policy, and some were not moved correctly.	Checked against the real value, corrected, and confirmed in a settings report.
Security records from staff computers never reached the management zone.	The firewall was missing an "allow" rule for them; it was added.
Record collection silently stopped after a restart.	The collecting computer started before the main server and misjudged its network. Its rule now works either way.
The firewall's control page used an unencrypted connection.	Switched to an encrypted connection.
The main server kept trying to reach the internet.	The firewall blocked every attempt, proving the server zone is sealed. The feature causing it was turned off.

Results

Tests from every zone confirmed each could reach only what it should. A ransomware exercise showed the game would stay online and the evidence would survive, but exposed two gaps: no automatic alerts, and backups stored on the same server they protect. Both are highly recommended for a real environment.